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Novel significant stage-specific differentially expressed genes in ulveal melanoma

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Abstract

Uveal melanoma (UM) is the most common primary intraocular malignancy in adults. The annual incidence in Europe and the USA is 6 per million population per year, with around 50% of patients developing metastatic disease, which usually involves the liver, and is usually fatal within 1 year (Jager *et al.* 2020b). Here, we performed an RNA-Seq expression analysis and visualization using data from patients diagnosed with UM. The goal is to compare the transcriptomic profile of patients in an early tumoural stage with ones in an advanced stage of this disease and find differential expressed genes in this type of cancer to shed some light into the mechanisms of the disease.

Keywords: GWAS; Ulveal Melanoma; R; SNP

Introduction

Uveal melanoma (UM), which arises from melanocytes of the choroid plexus, ciliary body, and iris of the eye, is the most common primary intraocular malignant tumour in adults. UM accounts for around 3% to 5% of all melanomas, with an incidence of roughly five cases per million in the USA. Up to 50% of patients develop metastases within 15 years of their initial diagnosis (National Cancer Institute (2006)); 90% of these tumours metastasize to the liver as the first site of metastasis, and a subset of patients also develop metastases to the lungs, bone or other organs. There are distinct mutational and expression profiles correlating with UM varying from case to case and tumour stage: mutations in *PLCB4* (encoding a phospholipase downstream of GNAQ/GNA11) and *CYSLTR2* (G-protein–coupled receptor) have been identified in previous studies (Jager *et al.* (2020b)).

Because of the shifting genomic landscape between cases, some treatments will get a better response than others. Identifying the differentially expressed genes between tumour stages underlying the illness is key for a complete understanding of UM and, therefore, an effective patient-oriented therapy.

RNA-Seq poses a great tool to analyse the differential expression between early and late stage tumours. RNA-Seq is a methodology for RNA profiling based on next-generation sequencing that enables the measurement and comparison of gene expression patterns with good resolution. The use of this technique will allow the identification of differentially expressed genes in a cohort of UM patients from different countries.

Materials and methods

Data procedence

The Cancer Genome Atlas (TCGA) is a collaboration between the National Cancer Institute (NCI) and the National Human Genome Research Institute (NHGRI) that has generated comprehensive, multi-dimensional maps of the key genomic changes in 33 types of cancer (National Cancer Institute (2006)). The TCGA dataset, comprising >2 petabytes of genomic data, has been made publicly available and is helping the research community to improve prevention, diagnosis and treatment of cancer.

Bioinformatic tools used

Recount 2 is an online resource consisting of RNA-Seq gene and exon counts for different studies, including TCGA data. The dataset was obtained from this source(Recount2 (2014)).

The code was run using the R Studio interface (RStudio Team (2020)), and finally, to format the work we used the text editor LaTeX.

The followed procedure was based on our previous study (Jornet *et al.* (2022)), first analizing the data frame and then running a code provided by our close collaborator Dr. Marta Coronado to generate graphical results. The packages used in the analysis include Bioconductor core packages for importing and processing raw sequencing data and loading gene annotations, namely SummarizedExperiment (M *et al.* (2021)), edgeR (MD *et al.* (2010)), DESeq2 (Love *et al.* (2014)), tweeDEseq (Esnaola *et al.* (2013)), tweeDEseqCountData (JR and M (2021)), GOstats

(Falcon and Gentleman (2007)), annotate (Gentleman (2021)), org.Hs.eg.db (Carlson (2019)), biomaRt (Durinck *et al.* (2009)) and ggplot2 (Wickham (2016)).

Data description

The RNA-Seq dataset comprehends a group of 80 individuals, 35 females and 44 males, born between 1926 and 1989 and diagnosed with UM. There was no data about tobacco and alcohol use, and BMI data was absent or incomplete. The number of genes in the analysis amounts to 58037. In order to proceed with the analysis, one patient was removed due to lack of tumour stage information. Of the remaining 79, 39 presented early stage tumour and 40, late stage tumour (Figure 1).

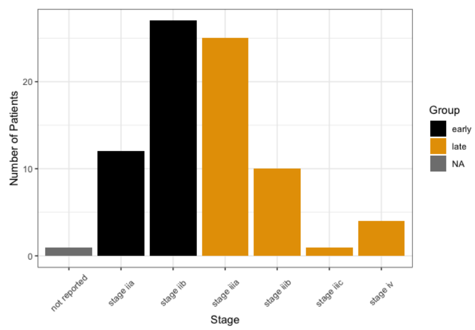


Figure 1 Early vs late tumour stage distribution across the studied cohort. Tumour stage related data is available for 79 of 80 patients.

Statistical analyses

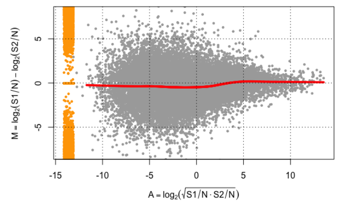
RNA-Seq data normalization . Comparison of gene expression between samples requires accounting for sequencing depth, transcript length and number of counts, which is proportional to the mRNA expression level. To ensure that technical bias has minimal impact on the results, we used the TMM method (Jager *et al.* (2020a)) and concluded that data did not require further normalization (Figure 2, 2a, 2b and 2c).

MA plots showcast if normalization is needed, where the logarithm of the ratio of level counts for each gene (M-values) is plotted against the log-average (A-values). Negative binomial distribution-based from DesEq R package allowed for testing differential gene expression between late and early subsets, keeping genes whose adjusted p-value was lower than 0.001, thus expecting genes overexpressed or underexpressed in late tumours vs early tumours.

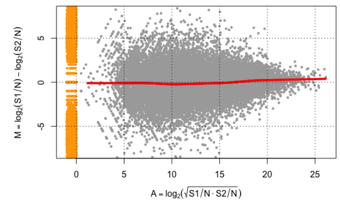
GO enrichment analysis . For the GO enrichment analysis, we used the `GOstats` R package in order to check association between GO terms and the candidate gene list.

Results and discussion

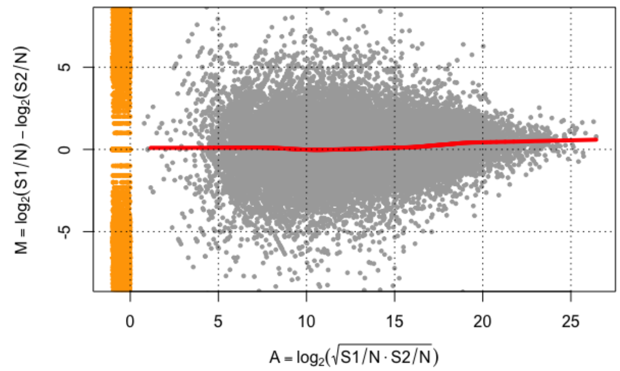
From the list of genes, we selected those with log-fold change >10 and adjusted p-value <0.001. Under these constraints, a total of 24 genes appear to be differentially expressed between early and late stage tumours, with 20 genes upregulated in late stage vs 4 genes downregulated in late stage. These differentially expressed genes are involved in various regulative pathways, with most of the genes involved in T cell regulation of tolerance induction and immune response (Figure 3, Figure 4 and Figure S6 in the Supplementary materials).



(a) RPKM normalization.



(b) TMM normalization



(c) Raw data.

Figure 2 MA plots of (a) data after PRKM, (b) data after TMM normalization and (c) raw data. A lowest fit (red line) is plotted underlying a possible trend in the bias related to the mean expression.

Uveal melanoma tumours show alterations in chromosomes 1, 3, 6 and 8. By far the most salient chromosomal aberration associated with metastatic uveal melanoma are chromosome 3 loss and 8q gain (Figure 5). The presence of chromosome 3 monosomy in a primary tumour strongly correlates with the risk of metastatic disease. The gain of region 8q, which is also associated with a reduced survival, occurs frequently in combination with chromosome 3 monosomy and is considered to be a later event induced by the loss of chromosome 3. Other chromosomal alterations such as 6q or 1p losses also augment the metastatic risk, while the gain of region 6p occurs almost in a mutually exclusive manner with chromosome 3 monosomy and is associated with a better prognosis (Coupland *et al.* (2020)).

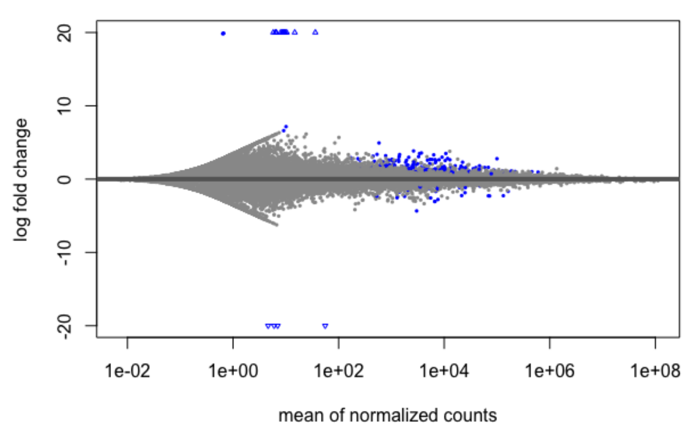


Figure 3 Log-fold change vs mean of normalized counts for all samples. Points are blue when the adjusted p-value is <0.1.

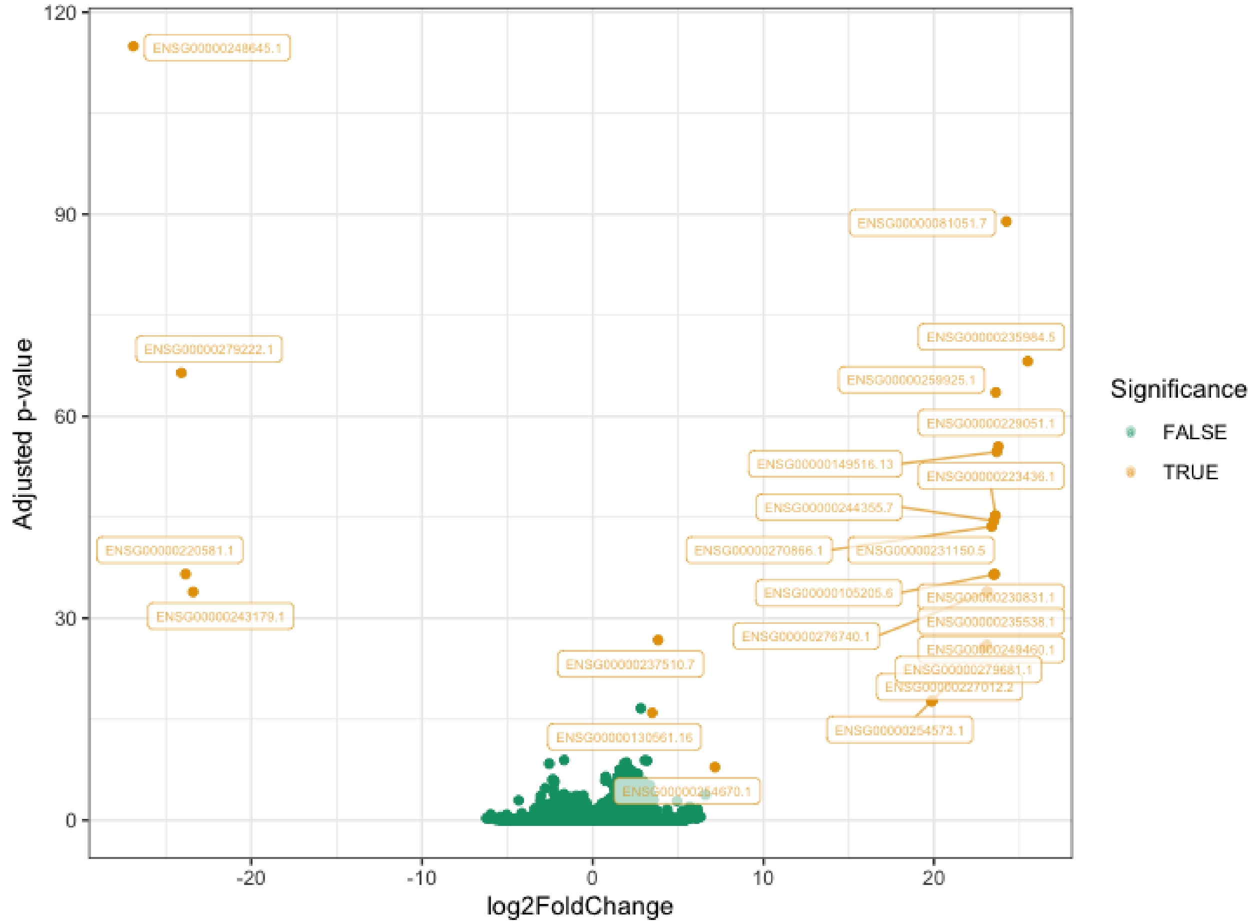


Figure 4 Scatter plot of `-log(p-value)` vs log2 fold change. Significant genes (adjusted `p-value < 0.001` and `log2 fold change > log2(10)`) are located towards the top of the plot and labbeled accordingly.

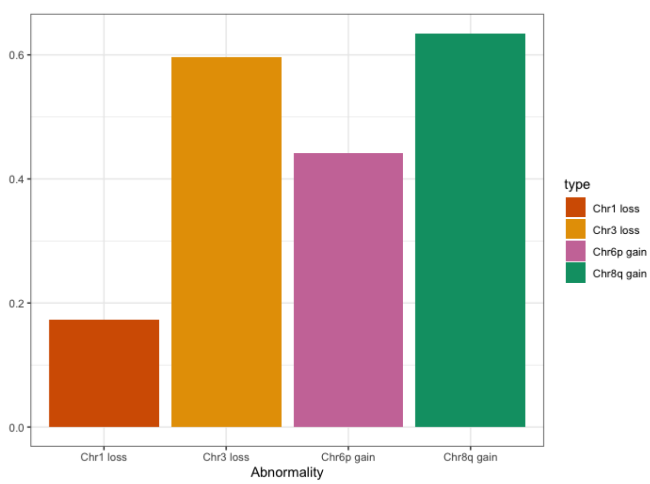


Figure 5 Distribution of chromosome abnormalities across the cohort. UM is characterized by chromosome 3 monosomy, which can induce 8q gain in later stages of the disease.

As with other types of cancer, UM malignancy is associated with mutations that influence T cell anergy. Several forms of T-cell anergy have been described, each occurring under specific conditions and in response to different stimuli (Abe and Macian (2013)).

Anergy has been reported to occur in T cells that infiltrate tumours and the immunosuppressive nature of the tumour microenvironment is thought to be responsible for this effect. In particular, the lack of dendritic cell maturation appears to account for an inefficient presentation of tumour antigens. In turn, suboptimally presented antigens, presumably due to the presence of inhibitory signaling and/or poor co-stimulation, would lead to the induction of T-cell anergy (Abe and Macian 2013).

Conclusions

With this study we proved that RNA-Seq is an efficient tool for differential expression analysis, showing that early and late stage tumours of UM present different genomic landscapes that may condition the development of the tumour and the illness.

Aknowledgements

We would like to thank, once again, Sir R. Fisher for his very useful annotations trough spiritual apparition, also the previous night of the publication, right when the authors were joyfully consuiming indecent amounts of ethanol at the town's chappel to evoke His always wise and truthful spirit. Due to business secrecy the authors can't provide the beverage's brand, but Jack Daniels et al. will be glad to privately shed light into the issue.

Funding

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Conflicts of interest

The authors declare no conflict of interest. Anyway, thank you Mr. Mc Donalds for your shinny coins. Keep producing such healthy foods! Bullshit, go vegan UwU.

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Supplementary materials

The figure resulting from the GO Enrichment analysis is provided following.

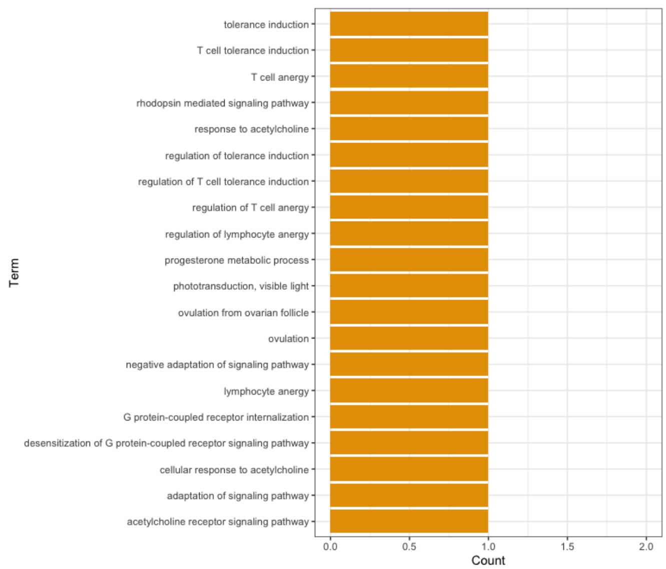

Figure S6 GO term enrichment analysis. Only the 20 terms with the smallest *p-value* are shown. Significance was set to *p-value*$< < 0.05$

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R code

In this section, the code to run the whole RNA-Seq analysis ispasted. The packages used are listed at the beginning. The code can also be found at the team's GitHub page https://github.com/andreasoler1234/R.Fisher-and-The-Tunas/blob/main/P4/RNASeq_Eye.R

```r
#####################################
#  RNA-Seq analysis for the UM case  #
#####################################

# CALLING required libraries
library(SummarizedExperiment)
library(edgeR)
library(DESeq2)
library(tweeDEseq)
library(tweeDEseqCountData)
library(GOstats)
library(annotate)
library(org.Hs.eg.db)
library(biomaRt)
library(ggplot2)
library(ggrepel)
library(stringr)

# SETTING THE WORK DIRECTORy
setwd(' ')

# READING THE DATA
setwd()
load(file = "data/eyeCancer/rse_gene_eye.Rdata")

# 1.RESHAPE THE DATAFRAMES ---------------------------------------------

#Add column `GROUP` to the meta-data to organize the cancer stages:
stage <- rse_gene$gdc_cases.diagnoses.tumor_stage

#id list of early and last tumours
ids.early <-grep(paste("stage i$", "stage ia$","stage ib$","stage ic$",
                       "stage ii$", "stage iia$","stage iib$",
                       "stage iic$",sep="|"), stage)

ids.late <-grep(paste("stage iii$", "stage iiia$", "stage iiib$",
                      "stage iiic$", "stage iv$", "stage iva$",
                      "stage ivb$", "stage ivc$",sep="|"), stage)


#Create an empty column named GROUP
colData(rse_gene)$GROUP <-rep(NA, ncol(rse_gene))
#Add early for those patients with tumours at stages i-ii
colData(rse_gene)$GROUP[ids.early] <- "early"
#Add late for those patients with tumours at stages iii-iv
colData(rse_gene)$GROUP[ids.late] <- "late"

dataGroups <- data.frame(Stage = rse_gene$gdc_cases.diagnoses.tumor_stage,
                         Group = rse_gene$GROUP)

#Visualization of early/late distribution (Figure 1)
ggplot(data = dataGroups, mapping = aes(x = Stage, fill = Group)) +
  geom_bar() + labs(y="Number of Patients") +
  scale_fill_manual(values=c("#000000", "#E69F00", "#56B4E9", "#009E73")) +
  theme(axis.text.x = element_text(angle = 45, vjust = 0.5, hjust=0.5)) +
  labs(title="Patients by stage")

#Check NA's and remove them
naData <- is.na(rse_gene$GROUP)
table(naData)
dim(rse_gene)
rse_gene <- rse_gene[, !naData]
dim(rse_gene)

#Summary of the patients
table(rse_gene$GROUP)

#Save the count data
counts <- assay(rse_gene, "counts")
counts[1:5, 1:2]

#Save the phenotype data
phenotype <- colData(rse_gene)
phenotype[1:5, 1:2]

#Check the order of the datasets
identical(colnames(counts), rownames(phenotype))    #TRUE

#Save the annotation data
annotation <- rowData(rse_gene)

# 2.RNA-SEQ DATA NORMALIZATION -------------------------------------------

#maPlor foe raw data (Figure 2a)
maPlot(counts[,1],
       counts[,2],
       pch=19,
       cex=.5,
       ylim=c(-8,8),
       allCol="darkgray",
       lowess=TRUE,
       xlab=expression(A == log[2] (sqrt(S1/N %.% S2/N))),
       ylab=expression(M == log[2](S1/N)-log[2](S2/N)))
grid(col="black")
title("Figure 1a")

#RPKM normalization method (figure 2b)
geneLength <- annotation$bp_length
counts.rpkm <- t(t(counts/geneLength*1000)/colSums(counts)*1e6)

counts.rpkm[1:5,1:2]

maPlot(counts.rpkm[,1],
       counts.rpkm[,2],
       pch=19,
       cex=.5,
       ylim=c(-8,8),
       allCol="darkgray",
       lowess=TRUE,
       xlab=expression( A==log[2] (sqrt(S1/N%.%S2/N))),
       ylab=expression(M==log[2](S1/N)-log[2](S2/N)))
grid(col="black")
title("Figure 1b")

# TMM normalization method (figure 2c)
counts.tmm <- normalizeCounts(counts, method = "TMM")

counts.tmm[1:5,1:2]

maPlot(counts.tmm[,1],
       counts.tmm[,2],
       pch=19, cex=.5,
       ylim=c(-8,8),
       allCol="darkgray",
       lowess=TRUE,
       xlab=expression( A==log[2] (sqrt(S1/N%.%S2/N))),
       ylab=expression(M==log[2](S1/N)-log[2](S2/N)))
grid(col="black")
title("Figure 2c")

# 3.RNA-SEQ RESULTS  -----------------------------------------------------

#Stage of each patient
pheno.stage <- subset(phenotype, select=GROUP)
```

```r
#Recreate the counts in a new matrix
counts.adj <- matrix((as.vector(as.integer(counts))), nrow=nrow(counts),
                     ncol=ncol(counts))

rownames(counts.adj) = rownames(counts)
colnames(counts.adj) = colnames(counts)

#Check information
identical(colnames(counts.adj), rownames(pheno.stage))

#Transform the group variable to factor
pheno.stage$GROUP <- as.factor(pheno.stage$GROUP)

#Create the DESeqDataSet input
DEs <- DESeqDataSetFromMatrix(countData = counts.adj,
                              colData = pheno.stage,
                              design = ~ GROUP)

#Differential expression analysis
dds <- DESeq(DEs)

#Extracts a result table from a DESEq analysis
res <- results(dds, pAdjustMethod = "fdr")

#Plot the differentially expressed genes in early vs late cancer (Figure 3)
plotMA(res, ylim=c(-20,20), main='10.4161/onci.22679')

#Find the genessignificanlty overexpressed or underexpressed in tumours
res.padj <- res[(res$padj < 0.001 & !is.na(res$padj)),]

#Keep genes that have a 10 log<sub>2</sub>fold-change.
res.padj_4fold <- res.padj[abs(res.padj$log2FoldChange) > log2(10),]
nrow(res.padj_4fold)

plotMA(res.padj_4fold, ylim=c(-10,10),
       main='Most differentially expressed genes')

#Create the dataframe with the results of the DE
resDF <- as.data.frame(res)
#New column with the gene names
resDF$gene <- rownames(resDF)
#New column representing if a gene is deferentially expressed or not
resDF$filter <- abs(res$log2FoldChange) > log2(10) & res$padj < 0.001
table(resDF$filter)

#Vulcano plot showing differential expression (Figure 4)
ggplot(subset(resDF, resDF$padj != "Na"),
  aes(x = log2FoldChange, y = -log(padj), colour = factor(filter))) +
  geom_point() + labs(y = "Adjusted p-value", x = "log2FoldChange") +
  geom_label_repel(data=resDF[abs(resDF$log2FoldChange) > log2(10) & resDF$padj < 0.001 ,], aes(label = as.factor(gene)), alpha = 0.7, size = 2, force = 1.3) +
  title("Adjusted P-value vs log2fold change") +
  scale_color_manual(values=c("#009E73", "#E69F00")) +
  labs(colour = "Significance")

#Save a list of DE genes, and we remove the last part of the id
deGenes <- gsub("\\..*", "",rownames(res.padj_4fold))

biomart_uses <- useMart(biomart="ensembl", dataset="hsapiens_gene_ensembl")
biomart_uses

Entrez_BM <- getBM(attributes=c("ensembl_gene_id", "entrezgene_id"),
                   filter="ensembl_gene_id",
                   values=deGenes,
                   mart=biomart_uses,
                   uniqueRows=TRUE)

dim(Entrez_BM)
head(Entrez_BM)

Entrez_ids <- as.character(na.omit(Entrez_BM$entrezgene_id))

universe <- mappedkeys(org.Hs.egGO)
count.mappedkeys(org.Hs.egGO)

# 4.GO ENRICHMENT ANALYSIS  ----------------------------------------------

#Set the threshold for the p value to 5%
GOtest <- new("GOHyperGParams",
              geneIds = Entrez_ids,
              universeGeneIds=universe,
              annotation = "org.Hs.eg.db",
              ontology = "BP",
              pvalueCutoff= 0.05,
              conditional = FALSE,
              testDirection = "over")

#Hypergeometric test
GOtestOver <- hyperGTest(GOtest)
GOtestOver

GOresult <- summary(GOtestOver)
head(GOresult)



#GO term enrichment result plotting (Figure S6)
cbPalette <- c("#E69F00", "#E69F00", "#E69F00", "#E69F00", "#E69F00",
               "#E69F00", "#E69F00", "#E69F00", "#E69F00", "#E69F00",
               "#E69F00", "#E69F00", "#E69F00", "#E69F00", "#E69F00",
               "#E69F00", "#E69F00", "#E69F00", "#E69F00", "#E69F00")

top10pvalterms <- head(GOresult,20)
ggplot(top10pvalterms, aes(x=`Term`, y=Count, label=Count, fill=Term)) +
  geom_bar(stat="identity") + coord_flip() + ylim(0, 2) +
  theme(legend.position = "none") + labs(title = "GO enrichment analysis (p value < 0.01)") + scale_fill_manual(values=cbPalette)

#Plot of chromosomal abnormalities across cohort (Figure 5)
ggplot(Chromosomal_abnormality,aes(x=type,y=Chr_abnormalities, fill=type))+
  geom_bar(stat = "identity") +
  labs(title = "Chromosomal abnormalities in ulveal cancer patients") + ylab(" ") + xlab("Abnormality")
  + scale_fill_manual(values=c("#D55E00", "#E69F00", "#CC79A7", "#009E73"))
```